Your endpoints have a firmware deadline – are you ready?

Secure Boot Certificate Updates

Agenda

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- What is Secure Boot?
- What happens after June 2026?
- Hands-on Remediation & Validation
- Preparing Your Environment for the Windows Secure Boot Certificate Update
- FAQ

Key takeaways:

- Understand what is expiring, why it matters, and what actually breaks if you miss the deadline
- Walk away with a realistic rollout plan and a short checklist you can use

Thank you Sponsors

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Microsoft

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Warm up

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Windows Secure Boot certificate expiration and CA updates

▸ *Applies To*

Original publish date: June 26, 2025

KB ID: 5062710

Change Log

Change Date	Change Description
February 3, 2026	▪ Added a new section "**The impact of Secure Boot certificate expiration**". ▪ Removed the notes marked as "**Important**" above and below the section "**Call to action**".
November 10, 2025	Corrected two typos under "New Certificate": ▪ from Microsoft Corporation KEK CA 2023" to "Microsoft Corporation KEK 2K CA 2023" ▪ and from "Microsoft Option ROM CA 2023" to "Microsoft Option ROM UEFI CA 2023"

Windows Secure Boot certificate expiration and CA updates - Microsoft Support

February 10, 2026—KB5077181 (OS Builds 26200.7840 and 26100.7840)

▸ *Applies To*

Release Date:	2/10/2026
Version:	OS Builds 26200.7840 and 26100.7840

Improvements

This security update contains fixes and quality improvements from KB5074109 (released January 13, 2026). KB5077744 (released January 17, 2026), and KB5078127 (released January 24, 2026). The following summary outlines key issues addressed by this update. Also, included are available new features. The bold text within the brackets indicates the item or area of the change.

- **[Gaming]** Fixed: This update addresses an issue that determines device eligibility for the full-screen gaming experience.
- **[Networking]** Fixed: This update addresses an issue that prevented some devices from connecting to certain WPA3-Personal Wi-Fi networks. The issue might occur after you install KB5074105.
- **[Secure Boot]** With this update, Windows quality updates include a broad set of targeting data that identifies devices and their ability to receive new Secure Boot certificates. Devices will receive the new certificates only after they show sufficient successful update signals, which helps ensures a safe and phased rollout.

February 10, 2026—KB5077181 (OS Builds 26200.7840 and 26100.7840) - Microsoft Support

Secure Boot Foundations

Core Concepts Revisited

What is Secure Boot?

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- UEFI firmware feature - ensures only trusted, digitally signed software runs at boot
- Validates digital signatures of boot components against trusted keys stored in firmware
- Industry standard defining protocols for certificate management, firmware authentication, and OS verification
- Based on a PKI model - Certificate Authorities (CAs) like Microsoft and OEMs issue and manage the trust certificates
- **Foundation: if a boot component isn't signed by a trusted CA, it doesn't run**

Threats Secure Boot helps prevent

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- **Bootkits and Rootkits**: Malware that attempts to compromise the bootloader or kernel during startup
- **Firmware Attacks**: Attempts to tamper with or replace UEFI firmware
- **Code Injection**: Unauthorized code injected into the boot process
- **Unauthorized Kernel Modules**: Loading of unsigned or tampered kernel modules
- **Exploitation of Boot-Time Vulnerabilities**: Attacks that target vulnerabilities in the boot sequence itself

Why are Secure boot Certificates expiring?

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- Introduced with Windows 8: Microsoft and OEMs provisioned devices with CAs defining what code is trusted at boot
- Certificates stored in device firmware - used to validate updates and boot components
- Original 2011 certificates are valid for 15 years → expiring in June 2026
- **After expiration: devices can no longer receive Secure Boot updates or pre-boot security fixes**
- **Result: devices remain stuck with 2011-era protections — vulnerable to new boot-level threats**

Windows Boot Sequence - Where is What?

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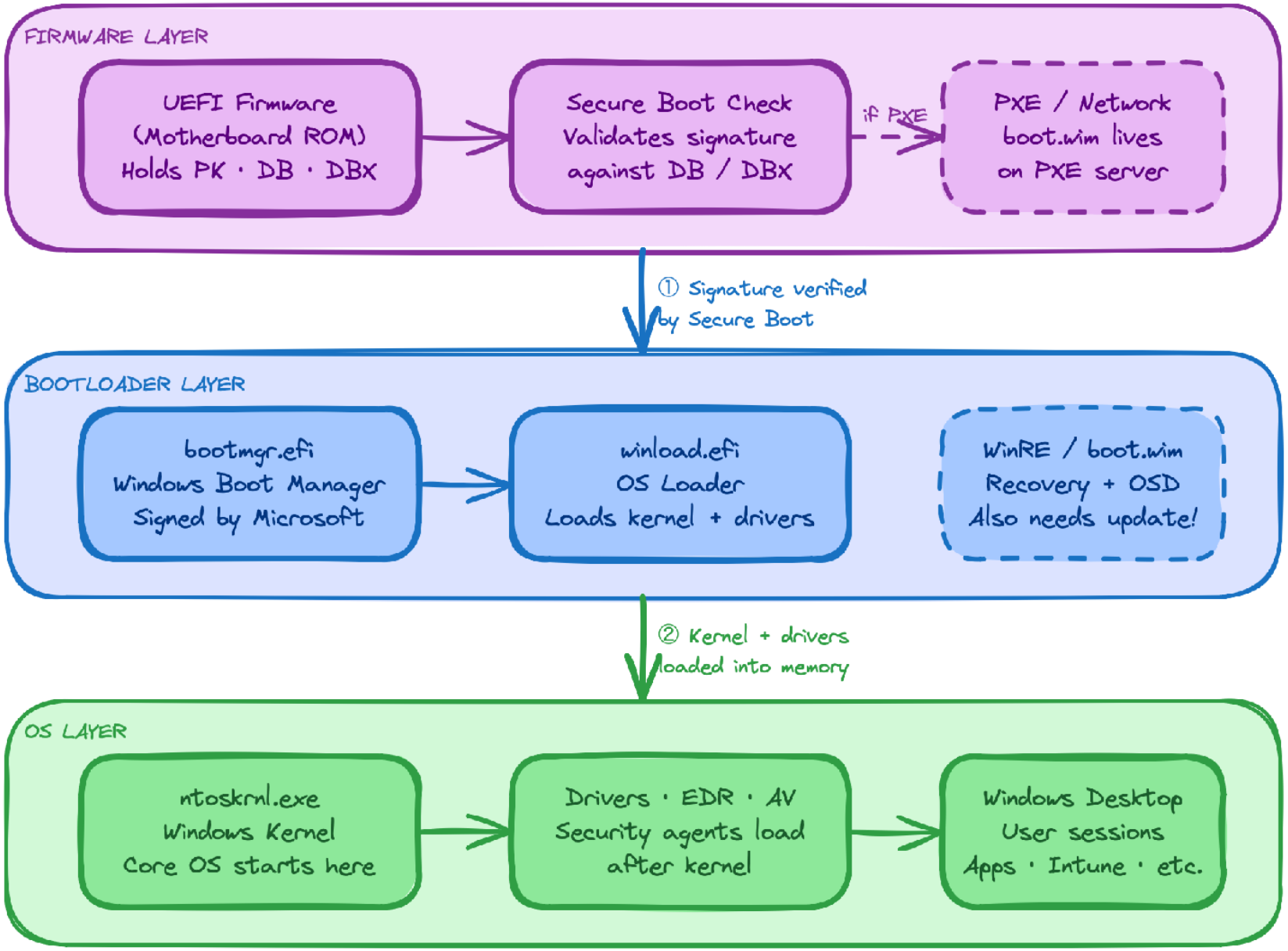

Secure Boot – Chain of Trust

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What’s changing?

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Expiring Certificate	Expiration date	New Certificate	Storing location	Purpose
Microsoft Corporation KEK CA 2011	June 2026	Microsoft Corporation KEK 2K CA 2023	Stored in KEK	Signs updates to DB and DBX.
Microsoft Windows Production PCA 2011	Oct 2026	Windows UEFI CA 2023	Stored in DB	Used for signing the Windows boot loader.
Microsoft UEFI CA 2011*	June 2026	Microsoft UEFI CA 2023	Stored in DB	Signs third-party boot loaders and EFI applications.
Microsoft UEFI CA 2011*	June 2026	Microsoft Option ROM UEFI CA 2023	Stored in DB	Signs third-party option ROMs

KEK: Key Enrollment Key **CA:** Certification Athority **DB:** Secure Boot Signature Database **DBX:** Secure Boot Revoked Signature Database

What happens if you do nothing?

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- **June 2026: Microsoft UEFI Secure Boot certificates expire**
- Secure Boot trust chain breaks - boot integrity protection weakened
- ~~BitLocker recovery prompts possible on unprepared devices~~
- Compliance dashboards turn red across your entire fleet
- **This is not hypothetical - Black Lotus already showed us the stakes**
- Devices may fail to boot after Windows Update applies new DBX entries

Black Lotus - Same Trust Chain

- **Black Lotus** (2022/2023): sophisticated UEFI bootkit - bypassed Secure Boot on fully patched Windows systems

- Exploited a **signed but vulnerable** Windows Boot Manager - trusted because the signature was valid

- Microsoft's response: **KB5025885** (July 2023) - updated DBX to revoke the vulnerable boot managers

- *What happened after this action?*

The Three Things You Must Remember

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1 Expiry does not mean instant failure

UEFI firmware does not check certificate expiry dates.
After June 2026, old-signed Boot Managers continue to boot normally.

2 The real deadline is the DBX update — date unknown

When Microsoft adds old signatures to DBX, old-signed Boot Managers are blocked permanently. No rollback once it arrives.

3 Act now — security is the reason, not just the deadline

An unupdated Secure Boot chain is an open attack surface — exactly as Black Lotus demonstrated in 2023.
Update the certificates because it is the right security posture.

⚠️ DBX update date not yet announced by Microsoft · Monitor: aka.ms/GetSecureBoot

Hands-on Remediation & Validation

Updating Microsoft UEFI Certificates (1/2)

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- **Status**: *Get-ItemProperty -Path HKLM:\SYSTEM\CurrentControlSet\Control\SecureBoot\Servicing\ -Name UEFICA2023Status | Select-Object UEFICA2023Status*

```
Administrator: PowerShell
PS C:\> Get-ItemProperty -Path HKLM:\SYSTEM\CurrentControlSet\Control\SecureBoot\Servicing\ -Name UEFICA2023Status | Sel
ect-Object UEFICA2023Status

UEFICA2023Status
----------------
NotStarted

PS C:\>
```

- **Update**: *Set-ItemProperty -Path 'HKLM:\SYSTEM\CurrentControlSet\Control\SecureBoot' -Name 'AvailableUpdates' -Value 0x5944*

  Start-ScheduledTask -TaskName '\Microsoft\Windows\PI\Secure-Boot-Update'

```
Administrator: PowerShell
PS C:\> Set-ItemProperty -Path
PS C:\> Start-ScheduledTask -T
PS C:\> Get-ItemProperty -Path
-Object UEFICA2023Status

UEFICA2023Status
----------------
InProgress

PS C:\>
```

```
Administrator: PowerShell
PS C:\> Start-ScheduledTask -TaskName '\Microsoft\Windows\PI\Secure-Boot-Update'
PS C:\> Get-ItemProperty -Path HKLM:\SYSTEM\CurrentControlSet\Control\SecureBoot\Servicing\ -Name UEFICA2023Status | Sel
ect-Object UEFICA2023Status

UEFICA2023Status
----------------
Updated

PS C:\>
```

Updating Microsoft UEFI Certificates (2/2)

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```
Administrator: PowerShell

PS C:\> Get-UEFICertificate.ps1 -Type DB

Type            : DB
Description     : Signature Database
Index           : 1
SignatureType   : a5c059a1-94e4-4aa7-87b5-ab155c2bf072
OwnerGuid       : 77fa9abd-0359-4d32-bd60-28f4e78f784b
CertificateSize : 1499
Subject         : CN=Microsoft Windows Production PCA 2011, O=Microsoft Corporation, L=Redmond, S=Washington, C=US
Issuer          : CN=Microsoft Root Certificate Authority 2010, O=Microsoft Corporation, L=Redmond, S=Washington, C=US
Thumbprint      : 580A6F4CC4E4B669B9EBDC1B2B3E087B80D0678D
Issued          : 10/19/2011 11:41:42 AM
Expires         : 10/19/2026 11:51:42 AM
SerialNumber    : 61077656000000000008
ParseError      :
RawData         : {48, 130, 5, 215…}

Type            : DB
Description     : Signature Database
Index           : 2
SignatureType   : a5c059a1-94e4-4aa7-87b5-ab155c2bf072
OwnerGuid       : 77fa9abd-0359-4d32-bd60-28f4e78f784b
CertificateSize : 1454
Subject         : CN=Windows UEFI CA 2023, O=Microsoft Corporation, C=US
Issuer          : CN=Microsoft Root Certificate Authority 2010, O=Microsoft Corporation, L=Redmond, S=Washington, C=US
Thumbprint      : 45A0FA32604773C82433C3B7D59E7466B3AC0C67
Issued          : 6/13/2023 11:58:29 AM
Expires         : 6/13/2035 12:08:29 PM
SerialNumber    : 330000001A888B980056228 4C100000000001A
ParseError      :
RawData         : {48, 130, 5, 170…}

PS C:\>
```

Updated Microsoft DB Certificates

Remediation options: choose your path

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- **Option A: Windows Update** - simplest, least control
  - Requires KB5012170 + current cumulative update; hard to control timing
- **Option B: Intune Remediation** - recommended for Intune-managed estates
  - Detect + Remediate pair; full logging; phased assignment via device groups
  - References: Florian Salzmann (scloud.work) and tbone.se two-part series
- **Option C: ConfigMgr Configuration Item** - best for ConfigMgr-managed estates
  - BIOS version check guards against premature remediation
- All three paths require BIOS firmware prerequisite check before cert update

The Ninja Playbook: Execute With Precision

Rollout Plan: 7 Phases to Full Compliance

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1. Assess

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- Inventory all endpoints: OEM, model, BIOS version, Secure Boot state
- Identify non-standard boot paths - dual boot, custom PK, PXE, air-gapped devices
- Check **UEFICA2023Status** registry key - value should ultimately read "*Updated*"
- Use Autopatch Secure Boot Status report if licensed (*and if it not revoked* 🤦)
- Sample Secure Boot Inventory Data Collection script

2. Prepare

- Confirm devices are on cumulative updates from ***February 13, 2024,*** or later - this is the minimum for UEFI CA 2023 enrollment

- Update OEM firmware proactively - firmware limitations can block the update entirely

- ~~Verify BitLocker recovery keys are escrowed in Entra ID or AD DS before any rollout~~

- ~~Validate that your helpdesk can retrieve recovery keys quickly if needed~~

3. Plan

- Define pilot group covering each major hardware and firmware combination
- Establish success criteria: normal boot, ~~no unexpected BitLocker recovery~~, PXE still works
- Decide: **auto-rollout or managed rollout** - Microsoft started automatic updates with KB5074109 (January 2026) for high-confidence devices
- Opt out with ***HighConfidenceOptOut = 1*** if you want explicit control (***telemetry basic/full***)

4. Deploy

- Set ***AvailableUpdates = 0x5944*** to deploy all certificates and update to 2023-signed Boot Manager

- Ring 0 - Pilot: IT team and representative hardware, manual validation
- Ring 1 - Broad: general population, monitored rollout
- Ring 2 - Critical: sensitive or high-availability systems, final wave

- Allow ***48 hours and one or more restarts*** after configuration change for full deployment

- ***Do not mix deployment methods*** on the same device

5. Media

- Rebuild WinRE images - Secure Boot trust changes affect recovery environment
- Update PXE boot images - ***bootmgr.efi*** must be signed with new certificate before DBX update rolls out
- Test PXE boot against pilot devices ***that have already received the DB update***
- Verify USB recovery media creation processes are updated

⚠ ***This step must happen before the DBX rollout - not after***

6. Monitor

- Event ID ***1808 = success -*** new certificates applied to firmware
- Event ID ***1801 = error -*** certificates not applied, investigate
- Registry key *UEFICA2023Status* = "Updated" → compliant
- Registry key *UEFICA2023Error* exists → deployment error pending
- Collect via Sentinel, Defender for Endpoint, Log Analytics or SIEM

7. Communicate

- End users require ***no notification*** - update runs silently, no user-facing prompts expected
- ***BitLocker suspension is not required*** - confirmed in Microsoft AMA March 2026, update does not trigger PCR changes
- ***Brief your helpdesk*** - if a recovery prompt occurs, they must know how to retrieve the key from Entra ID or AD DS
- ***Pilot ring first*** - validate BitLocker behavior before broad rollout; if prompts occur, adjust communication accordingly

Useful links

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- Registry key updates for Secure Boot: Windows devices with IT-managed updates
- Windows Secure Boot certificate expiration and CA updates
- Microsoft Intune method of Secure Boot for Windows devices with IT-managed updates

- Dell
- HP
- Lenovo

- Get-UEFICertificate PowerShell Script on GitHub
- Omnissa Community

Virtual Servers Are Not Exempt

Emulated UEFI firmware requires the same certificate maintenance as physical hardware

Every Platform Is Affected

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Topic	VMware	Hyper-V	Azure VMs	Windows 365	AVD
Secure Boot enabled?	Check per VM	Gen 2 = yes	Trusted Launch	Yes	Yes
UEFI Firmware Update	ESXi ⩾ 8.0.2	WS Updates	Automatic	Automatic	Automatic
PK/KEK/DB Update	Known bug < 8.0.2	Via WU	Automatic	Automatic	Check custom images
Audit Tool	Get-VMFirmware	Get-VMFirmware	Azure Policy	MS managed	MS managed

VMware / Broadcom

- VMs use **OVMF** (Open Virtual Machine Firmware) - an emulated UEFI with its own certificate store, independent from the host OS
- **ESXi < 8.0.2**: known Platform Key (PK) bug - PK is NULL or incorrectly set, blocking all KEK and DB updates entirely
- VMs migrated from older ESXi hosts **inherit the PK bug** - even if the current host is patched, the VM firmware state does not automatically correct itself
- ***Action**: audit ESXi versions and VM hardware versions first, identify VMs with NULL Platform Key as a separate remediation track before any certificate rollout*

Hyper-V

- **Generation 2 VMs only** - Generation 1 VMs have no UEFI and no Secure Boot, certificate updates are not applicable
- Secure Boot certificates in virtual firmware are updated via **Windows Server cumulative updates** - same mechanism as physical devices
- **Windows Server 2019 and older**: automatic certificate updates are not guaranteed - verify manually and consider managed deployment via registry key or Group Policy
- ***Action**: audit all Gen 2 VMs for Secure Boot status and template type – **Get-VMFirmware** shows SecureBoot state and the active SecureBootTemplate per VM*

Azure VMs

- **Trusted Launch VMs**: Microsoft manages the virtual UEFI firmware update process automatically - no manual intervention required

- **Legacy Azure VMs** without Trusted Launch have no Secure Boot enabled - they are outside the certificate update scope but also outside boot-level protection entirely

- On-premises VMs replicated to Azure (Azure Site Recovery) carry their **source firmware state** - a VM with a broken PK from VMware remains broken in Azure

- ***Action**: use Azure Policy to identify VMs without Trusted Launch, treat them as a separate security risk conversation alongside the certificate rollout*

FAQ

- AMA from Microsoft
- Server
  - Windows Server Secure Boot playbook for certificates expiring in 2026
- Mini Course – Secure Boot 2026 - What Breaks and How to Fix It – 02.04.2026

Don’t be (lazy) Ákos

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QUINTESSENZ · KEY TAKEAWAY

You cannot *skip the stairs.*

- **Secure Boot is not advanced security** — it is the foundation every higher control depends on
- **June 2026 is not a new problem** — it is deferred maintenance becoming a hard deadline
- **Black Lotus proved it** — an untrusted boot chain defeats every control above it
- **Start this week.** Assess first. The deadline is closer than it looks.

> *"You cannot build a secure environment on an untrusted boot chain. Fix the foundation first."*

Thank you